

Monte Carlo Simulation for Financial Analysis

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Abstract

Monte Carlo Simulations are simulations of a certain random event. What makes them notable is that they involve running these random events thousands of times. This creates a set of events that all come from a similar beginning state but eventually change completely due to the randomness of the probability function they come from.

Monte Carlo Simulations have a strong place in finance, due to the fact that finance is extremely difficult to predict, and seemingly random. Investments are run from the equation:

$$I_{\text{final}} = I_{\text{initial}}(1 + r)$$

Where r is the return over the period from the initial investment to the final one, and I is the investment. Returns can be modeled by certain probability functions or other method, and after being run hundreds or thousands of times, unique investment outcomes are created.

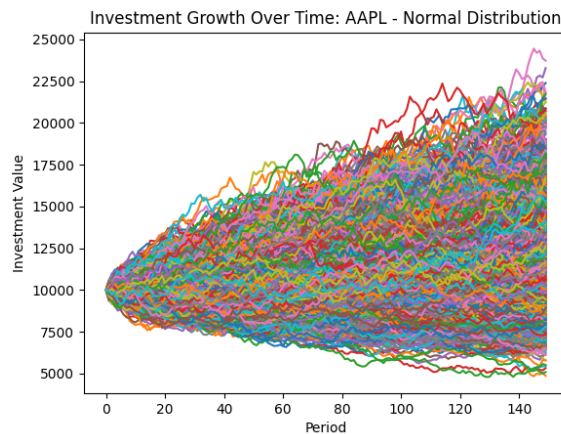


Figure 1: Monte Carlo Simulation of Investments

1 Overview

This project involves creating a Monte Carlo Simulation using returns from a normal distribution and historical bootstrapping. The goal of this project was to use a basic financial model to interpret and analyze a financial situation. As this is my first project like this, I wanted to go in depth into this basic simulation and its results, which is why no trading strategy was used to make things more complicated. Financial Data from the last 10 years from Apple, Tesla, and Microsoft was downloaded from Yahoo Finance for this project and used to make the models.

This model is extremely basic, as there are many assumptions made to just focus on the simulation of stock prices. No dividends, interest rates, inflation, or taxes were added to the simulation. While these are a large part of financial decision making, for this project, they were not deemed necessary and would only complicate things.

From this simulation, the biggest findings were the similarities and differences of the normal and historical bootstrapping methods, and the analysis of risk in the model. The normal and historical bootstrapping models provide very similar results to the Monte Carlo Simulation, with some differences being less skewness on the normal distribution model due to it preserving the general shape of the data, and providing less outlier results. For risk analysis, the Value at Risk and Conditional Value at Risk at a 99% confidence were calculated. It was found that these values were very high, high enough for someone to fear them.

Important takeaways from this project are that investing in companies without a clear strategy and goal will not work out very well. The proportions of loss for this model were high, around 20-30%, and a majority of investments in the simulation made little to no money. It is important to note that Tesla's investments were much more variable compared to Microsoft and Apple's. Tesla's final investments were highly skewed right, and the volatility in final investment price was extremely high.

2 Simulation

The Monte Carlo Simulation was implemented by python code in VS Code. The simulation begins with a single investment. This investment is repetition of the equation

$$I_{\text{final}} = I_{\text{initial}}(1 + r)$$

Over a certain number of periods, which in this simulation, the periods represent days. What this means is that every day, The investment is multiplied by the rate of return. Over n days, the investment becomes:

$$I_{\text{final}} = I_{\text{initial}}(1 + r_1)(1 + r_2) \dots (1 + r_{n-1})$$

Which is

$$I_{\text{final}} = I_{\text{initial}} \prod_{i=1}^{n-1} (1 + r_i)$$

While the recursive definition seems more straightforward and useful, defining the investment as a product does become useful, as shown later.

Now the only part left of the equation is the returns. The investment can be anything, but the returns cannot. Inputting in completely random return numbers is nonsensical, they must come from something. There are many different strategies to define returns. This project will use 2:

1. Returns from the Normal Distribution
2. Returns from Historical Bootstrapping

2.1 Returns from the Normal Distribution

The Normal Distribution is a probability distribution defined from negative to positive infinity, centered at a mean, μ and stretched by the standard deviation, σ ¹. In Finance, the normal distribution can be defined with expected return as the mean and volatility as the standard deviation.

Volatility is how much the stock price changes. This measurement of fluctuation is extremely important, as being able to understand how stable a stock price is one of the major parts in making smart decisions about investments.

Expected Return is explained by the name, it is the expected rate the stock will increase or decrease.

From this, it makes sense that Expected Return and Volatility become the mean and standard deviation, as the Mean is the average and the standard deviation is how much the data deviated from that average.

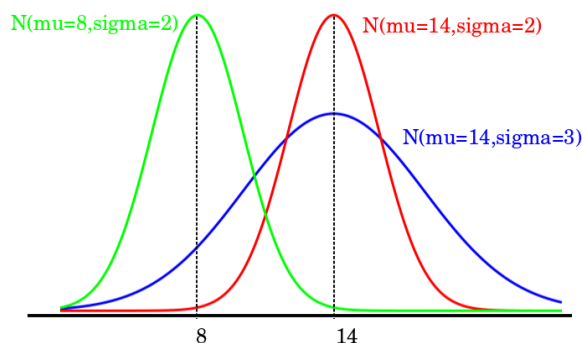


Figure 2: Transformations of the Normal Distribution

$N(\mu, \sigma)$ describes the Normal Function, with expected return and volatility, and this can be used to generate random numbers around that expected return with deviation. These random numbers can become the set of returns R , where each r_i is generated by a this random number generator and used in the equation for the final investment.

But how are Expected Return and Volatility found? Once again, inputting random numbers into expected return and volatility is illogical. Using real financial data, the expected return and volatility can be found. Using a set of returns taken from Yahoo Finance over the a certain period, taking the mean and standard deviation of this data set gives the expected return and standard deviation, respectively.

Now, Finally, the Monte Carlo Simulation is ready to be computed. Repeating the equation for investments over and over again, a set of final investments are created.

2.2 Returns from Historical Bootstrapping

Historical Bootstrapping is a method of testing using data from a certain period. This method does not involve the calculations of Expected Return and Volatility, only a set of returns. The data for this comes from Yahoo Finance as well.

The idea of Historical Bootstrapping is that, instead of using a model to generate returns, the returns are history from a period of time in which all returns are listed, for example, importing return data for each day

¹The Normal Distribution can also take another input in place of standard deviation: Variance, which is σ^2 . Because variance and standard deviation are directly related, for most purposes, it doesn't matter which it is defined by. For this financial analysis, Standard Deviation will be used.

from January 1st, 2025 to January 1st 2026. These returns can be used for any set of time. From this set R , returns can be randomly chosen to create a separate set R_{period} where $|R_{period}| = n$, where n is the number of periods (Again, for this project, days)

Now, instead of having a model generate returns, these returns are purely historical and can be used to calculate a final investment, and with the Monte Carlo simulation, numerous final investments.

These two methods are how investments over a number of days, was found, and the results will be shown, explained, and analyzed.

Each simulation produced with these 2 methods are inherently similar to every other simulation using the same data and same method. While a simulation might give slightly different statistics, the numbers will never be so different that a completely different result is created. The statistics that will be discussed in the next few pages, like proportion of losses and average final investment, only differ by a small amount. When comparing these values, 1 simulation will be used to minimize difference, but these simulations are all very similar.

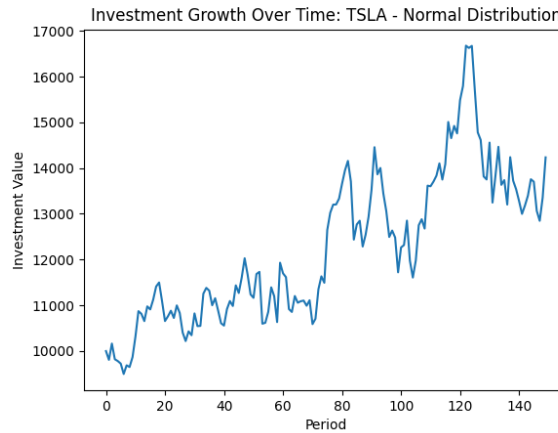


Figure 3: Singular Investment from the Normal Distribution

3 Graphical Analysis

Now that the 2 methods for determining returns are properly defined, Implementation can begin. For this project, 3 different sets of financial data were used, all taken from Yahoo Finance. The companies used were Apple (AAPL), Microsoft (MSFT)², and Tesla (TSLA). Daily returns from the past 10 years were used. For the Normal Distribution Monte Carlo Simulation, 3000 trials were repeated. For the Bootstrapping Monte Carlo Simulation, 5000 trials were repeated. Note that it does not necessarily matter that a different number of trials were recorded, as both simulations are inherently random and there are *enough*³ of them. For all simulations, \$10000 was invested into each company, over 150 days. No trading strategy was used, the money was just left for 150 days.

²No graphs for the Microsoft investments were deemed necessary, but information on its simulations will be referenced.

³Enough can mean many things in the subject of financial statistics. Sometimes there is a certain number that is the baseline for enough. In this case, enough means that changing the number of trials run by a small margin (Say ten) is negligible to the outcome of the simulation.

The simulations can be plotted in a few ways, some of which were already shown. A line graph of all investments can be made, and while this might look appealing and help visualize specifically what the Monte Carlo Simulation does, this graph does not hold very much weight quantitatively.

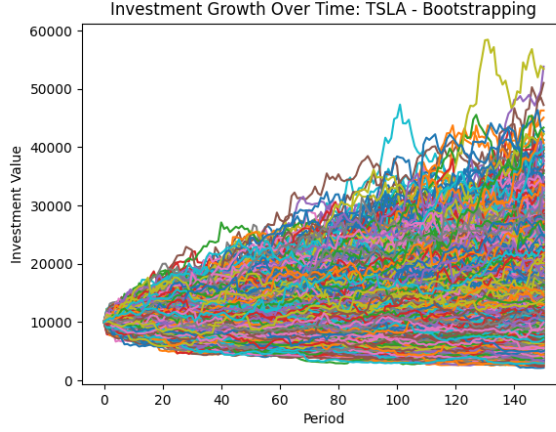


Figure 4: Line Graph Tesla Investment with Bootstrapping

3.1 Proportion of Losses

The proportion of losses for all investments is calculated with each simulation, let $\hat{g}$ be the proportions of loss of the normal simulations, and $\hat{b}$ be the proportions of loss from the bootstrapping simulations.

$$\hat{g}_{AAPL} = 0.232 \quad \hat{g}_{MSFT} = 0.269 \quad \hat{g}_{TSLA} = 0.334$$

$$\hat{b}_{AAPL} = 0.234 \quad \hat{b}_{MSFT} = 0.269 \quad \hat{b}_{TSLA} = 0.337$$

Next, using the confidence interval formula, six 95% confidence intervals can be created.

$$\text{C.I.} = \hat{p} \pm 1.96 \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

$$\text{C.I.}(g)_{AAPL} = [0.217, 0.247] \quad \text{C.I.}(g)_{MSFT} = [0.253, 0.285] \quad \text{C.I.}(g)_{TSLA} = [0.317, 0.351]$$

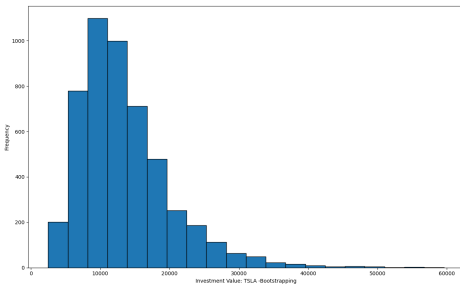
$$\text{C.I.}(b)_{AAPL} = [0.219, 0.249] \quad \text{C.I.}(b)_{MSFT} = [0.257, 0.281] \quad \text{C.I.}(b)_{TSLA} = [0.324, 0.350]$$

The confidence intervals for each company are extremely similar, giving reason to believe that the bootstrapping and normal distribution methods give fairly similar results when predicting proportions of loss. Not only that, but Microsoft's and Tesla's bootstrapping confidence interval is contained in its normal distribution confidence interval, meaning that bootstrapping provides less variability in loss. The Apple confidence intervals are very similar as well, but are the same size.

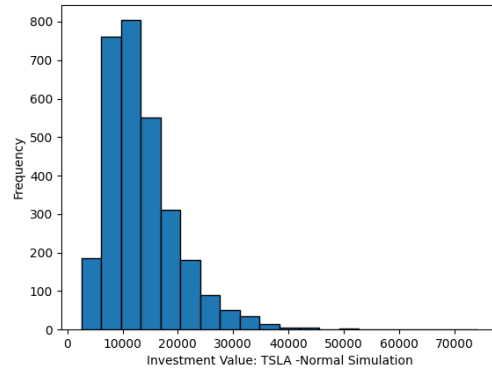
These proportions of loss are somewhat high, if this was real money, any investor would think twice about these investments. Also, if you increased the bar, instead of just doing proportion lost making it proportion that made less than 1000 in profit, the proportion would become much greater. A person's odds of this investment being worthwhile is not high at all.

3.2 Histograms

Histograms are much more useful for determining differences between Bootstrapping and the Normal Distribution than the line graphs, and they are also very helpful in determining the outcomes of the investment.

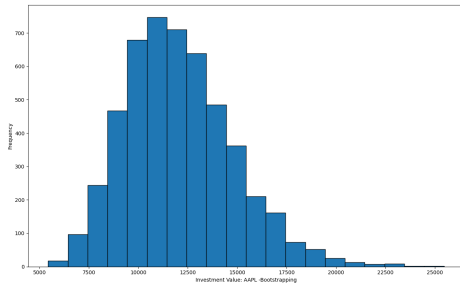


(a) Histogram of Tesla Investment with Bootstrapping

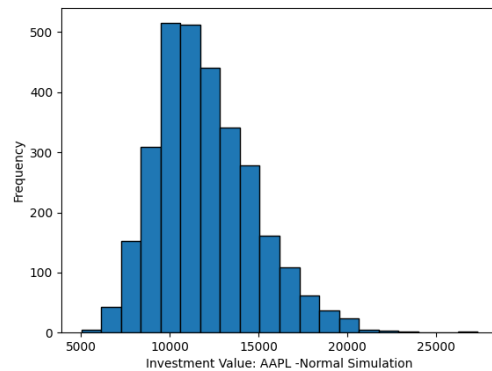


(b) Histogram of Tesla Investment Normal Distribution

Figure 5: Comparison of Bootstrapping and Normal Distribution Histograms for Tesla



(a) Histogram of Apple Investment with Bootstrapping



(b) Histogram of Apple Investment Normal Distribution

Figure 6: Comparison of Bootstrapping and Normal Distribution Histograms for Apple

From these histograms, it looks like for both companies, bootstrapping and the normal distribution provide fairly similar results. Both of their ranges are similar. However, at each peak, the counts are fairly different. Another thing to notice, especially about Tesla, is that its histogram of final investments is fairly skewed right. This means that final investments are not normally distributed. This makes sense, as the final investments come from the formula:

$$I_{\text{final}} = I_{\text{initial}} \prod_{i=1}^{n-1} (1 + r_i)$$

For the normal distribution, this means that they are the product of normally distributed values, and the normal distribution does not hold over multiplication. This distribution is *Lognormal*. By taking the natural log of the equation for final investment, what you get is a series, and it is known that addition holds the normal distribution, meaning that the natural log of final investments is normally distributed.

$$\begin{aligned} \ln(I_{\text{final}}) &= \ln(I_{\text{initial}} \prod_{i=1}^{n-1} (1 + r_i)) \\ \ln(I_{\text{final}}) &= \ln(I_{\text{initial}}) + \ln(1 + r_1) + \dots + \ln(1 + r_{n-1}) \\ \ln(I_{\text{final}}) &= \ln(I_{\text{initial}}) + \sum_{i=1}^{n-1} \ln(1 + r_i) \end{aligned}$$

3.3 Linear Combinations of Normal Distributions

Note: This section is more theoretical than applicable. Although there is a connection to financial analysis, it is not a large one⁴.

Looking a bit more at the normal distribution, it is known that the normal distribution is closed under addition and not multiplication, meaning that adding 2 normal distributions together creates another normal distribution. Does this mean that a linear combination of normally distributed random variables, like returns, is a normal distribution?

Let $R_1 \sim N(0, 1)$, where R_1 is a random variable representing returns that are normally distributed, like in the Normal Distribution Simulation.

Let $R_2 = aR_1 + b$, where $a, b \in \mathbb{R}$ such that R_2 is a linear transformation of R_1 .

The distribution of this linear transformation is given by the equation:

$$f_{R_2}(r_2) = \frac{f_{R_1}(h^{-1}(r_2))}{|h'(h^{-1}(r_2))|}$$

Where f_R is the probability density function of the respective variable, and $h(r_1) = r_2$

This means that $h(r_1) = ar_1 + b$, and $f(r_1) = \phi(r_1)$, where ϕ represents the normal probability density function. This is the function that computes the randomized returns in the simulation. From this $h^{-1}(r_2) = \frac{r_2 - b}{a}$, and $h'(r_1) = a$

Plugging these into the equation, this means that

$$f_{R_2}(r_2) = \frac{\phi(\frac{r_2 - b}{a})}{|a|}$$

And since

$$\phi(x, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

⁴At least at my current knowledge.

This means that a must be the standard deviation, and b must be the mean, or $R_2 \sim N(b, a)$. This shows the linear transformation of the Normal Distribution, and the fact that a linear combination of a Normal distribution is normally distributed.

What about $R_n = a_1R_1 + b_1 + a_2R_2 + b_2 + \dots + a_{n-1}R_{n-1} + b_{n-1}$? This question can be worded differently.

Is the set of all Normal Distributions $\mathcal{N}(\mu, \sigma)$ such that $\mu \in \mathbb{R}$, $\sigma > 0$, $\in \mathbb{R}$ a vector space? A vector space implies that linear combinations work in perfect order⁵ and are contained in the space, which is what the question is implying.

Suppose $a > 0$, Let $R_{n+1} = a_1R_1 + b_1$. This means R_{n+1} is normally distributed. Then, let $R_{n+2} = a_2R_2 + b_2$, and R_{n+2} is normally distributed. Continuing this for all random variables, their sums can be added together as well because they are themselves linear combinations, so their sums will be normally distributed, making R_n normally distributed. Note that any real number can be expressed as the sum of infinitely many real numbers, so the number of constants and scalars do not have to be equal.

However, suppose that $R_3 = R_2 - R_1$. Where both input variables are normally distributed. It is known that $\sigma_{X+Y}^2 = \sqrt{\sigma_X^2 + \sigma_Y^2} = \sigma_{X-Y}^2$. What this means is that when you subtract 2 normal distributions, the outcome of its standard deviation is not a linear relationship, meaning that the difference of two normal distributions is not normal distributed. What this means is that $\mathcal{N}(\mu, \sigma)$ is not a vector space.

What does this mean for the financial analysis? Returns are a percentage and never negative, so the scalar will never be negative. So while $\mathcal{N}(\mu, \sigma)$ is not a vector space, The natural log of the returns is somewhat closed under scalar multiplication and vector addition, and this will be a normal distribution, because returns are never negative, and neither is the initial investment. The Lognormal distribution of final investments has a normal distribution associated with it, shown by the equation shown on page 6. While returns are hardly ever normally distributed, treating them like they are provides a simulation that is easily understood mathematically, and compared with the bootstrapping simulation, they are not too different, showing the normal simulation works for modeling investment.

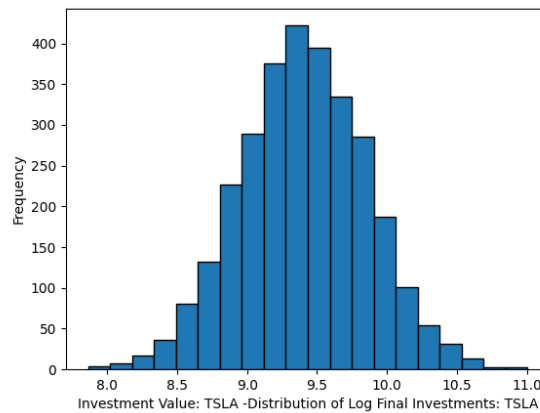


Figure 7: Distribution of Log Final Tesla Investments

⁵This is not the rigorous definition of a vector space, but it is what the idea of a vector space is built upon.

3.4 Time Fluctuation

Another area to look at regarding the final investments is the Mean and Standard Deviation of the final investments. These can be calculated with the simulations. The average final investment for all 3 companies hovers around 13000. A confidence interval could be created to describe this average, but it is not necessary. The standard deviation of the final investment varies heavily depending on the company. Looking back at figures 5 and 6, Tesla's standard deviation is extremely large, shown by the skewness. Microsoft and Apple's Standard deviations are not exactly the same, but very similar compared to Tesla's standard deviation. The reason for these differences in standard deviations but not averages comes directly from the source: The returns. The calculated volatility from Tesla's returns over the past 10 years is much higher, and this influences the standard deviation of final investment to be very large.

The reason for this discrepancy could be because of the large time interval that the data is taken from (10 years) along with each companies growth throughout this time. Tesla has arguably undergone the largest amount of growth during this period, leading to more fluctuating returns, affecting the volatility. Apple and Microsoft were both cemented companies by 2016, and Tesla was just starting to explode.

Another reason could come from the fact that Tesla experienced an extremely fast shoot up, and has somewhat hovered at the same stock price for the past few years, shooting down at times and then shooting back up. Apple and Microsoft have been steadily increasing over the 10 years period.



Figure 8: Tesla Stock Prices

The period the data is collected from is not the only evidence of times effect on this investment. Changing the amount of time the investment is left for also affects the investment. Obviously, if the investment is left for longer there is a greater chance of it making more money, because it has more time to mature, but there is also a greater chance of it losing more money. The longer the investment sits for, the more that can happen. This becomes a lesson about trading: Expecting \$10000 profit after a month is highly highly unlikely, unless the investment is extremely large. A tiny investment, say \$1000 would not generate much profit, but it is possible to profit a lot off of this using the right methods. Day Trading also does not generate a lot of money, but again, with the right methods it can be beneficial. At the other end of the spectrum, large losses can become extremely common because of time. Multiple compounding investments can lead to large changes extremely quickly.



Figure 9: A Financial Crash

4 Risk Analysis

Risk analysis is deciding what to do about investments regarding loss, and, like the name, risk. From the proportion of losses shown before, you can see that the odds of these investments being worthwhile is not very high. Some other statistics can be analyzed to measure risk as well.

4.1 Value at Risk

Value at Risk, often called VaR, is way to measure how much you would expect⁶ to lose in the worst case scenario from an investment. Value at Risk is a confidence interval, meaning we must choose a proper α level to be the percentage of confidence you could expect this occur.

In order to calculate Value at Risk, the losses must be defined as:

$$\text{Loss} = I_{\text{final}} - I_{\text{initial}}$$

Where a loss is when this is less than 0, and a profit is when this is above 0.

These differences (Meaning losses, profits, and neithers) Must be sorted into a list from lowest to highest, where the bottom $\alpha\%$ is the interval of losses one could expect at this confidence interval. The Value at Risk is the upper bound of this interval, meaning that a person is $(1 - \alpha)\%$ confident that the loss will be less than this value. Alternatively, there is a $\alpha\%$ chance that the maximum loss will be larger than the value at risk. This defines the value at risk as:

$$VaR = I(n\alpha)_{\text{final}} - I_{\text{initial}}$$

Where $I(n\alpha)$ is the final investment at the bottom α percentile.

For this simulation, the Value at Risk will be calculated for Microsoft, Tesla, and Apple, for both the Normal Distribution and Bootstrapping Simulations. Again, g represents the normal distribution simulation and b represents the bootstrapping simulation.

$$VaR(g)_{AAPL} = 2984 \quad VaR(g)_{MSFT} = 2924 \quad VaR(g)_{TSLA} = 5839$$

$$VaR(b)_{AAPL} = 2965 \quad VaR(b)_{MSFT} = 3118 \quad VaR(b)_{TSLA} = 5717$$

For all of these values at risks, the confidence level is 99%, meaning that there is a 99% chance that for investing in these companies, you will not lose more than these values. However, these values are not very low, which is not good for any investor. Risking almost \$6000 investing in Tesla would most likely not be worth it.

4.2 Conditional Value at Risk

Conditional Value at Risk, often called CVaR⁷, is a statistic that is somewhat similar to Value at Risk and used to quantify risk. Now, instead of creating a confidence interval for maximum expected loss (VaR), Conditional Value at Risk is actually the expected loss inside that confidence interval.

In probability theory, Expectation is defined as: $E(x) = \sum_i x_i p_X(x_i)$ So for the expected loss, $x_i = L(i)$ and $p_X(x_i) = \frac{1}{l}$. This gives

$$CVaR = \sum_i \frac{L(i)}{l}$$

⁶Kind of. While Conditional Value at Risk is mathematically what you would expect to lose, Value at Risk also is somewhat, although it is an upper bound to loss at a certain confidence level.

⁷And even Expected Shortfall.

Where $L(i)$ is loss i and l is the number of losses⁸ in the confidence interval. Now that Conditional Value at Risk is defined, it can be computed for all 6 possible simulations.

$$\text{CVaR}(g)_{AAPL} = 3512 \quad \text{CVaR}(g)_{MSFT} = 3411 \quad \text{CVaR}(g)_{TSLA} = 6372$$

$$\text{CVaR}(b)_{AAPL} = 3511 \quad \text{CVaR}(b)_{MSFT} = 3597 \quad \text{CVaR}(b)_{TSLA} = 6372$$

Once again it is shown that Teslas investment is at much higher risk than Apples or Microsofts. This type on information on risk analysis could influence a person not to invest in Tesla and take their money elsewhere. What this also shows once again is that The bootstrapping and Normal simulations are similar. The Conditional Values at Risk are also using the 99% Confidence interval, and these can be interpreted as the expected loss in the bottom 1% scenarios.

Conditional Value at Risk is important because it does not just show the barrier of loss, like Value at Risk. It is a more pessimistic, or better risk controlling value because it is an expected value.

For both Conditional Value at Risk and Value at Risk, a confidence interval would create a better understanding of the values, as this is from 1 simulation, however it is not a necessary computation for this project. These risk statistics create enough of a understanding of what the investments would do at maximum loss.

4.3 Confidence Levels

If the α was increased, what would happen?

Alpha is responsible for the width of the confidence interval, and making α lower increases the confidence interval, increasing the Value at Risk and Conditional Value at Risk. This idea comes from the basic statistics of the confidence interval and what α intrinsically means.

The value at risk tends towards 0 when α approaches the proportion of loss. For every simulation, the proportion of loss is different and the Value at Risk will only be 0 if $\alpha = p_{\text{loss}}$, this can be generalized using the confidence intervals for the true proportion lost created previously. If α is inside any of these confidence levels for the simulation, then the Value at Risk will be very close to 0. There is 95% confidence that the true proportion loss is inside these intervals.

$$\text{C.I.}(g)_{AAPL} = [0.217, 0.247] \quad \text{C.I.}(g)_{MSFT} = [0.253, 0.285] \quad \text{C.I.}(g)_{TSLA} = [0.317, 0.351]$$

$$\text{C.I.}(b)_{AAPL} = [0.219, 0.249] \quad \text{C.I.}(b)_{MSFT} = [0.257, 0.281] \quad \text{C.I.}(b)_{TSLA} = [0.324, 0.350]$$

For Conditional Value at Risk, changing α changes the size of the set of all values less than the Value at Risk. What this means is that more values contribute to the Conditional Value at Risk, meaning that $\text{CVaR} \propto -\text{VaR}$.

There is not a definitive relationship between CVaR and VaR besides the negative one. This is because they both come from random variables, so there is no function to describe them.

⁸Losses can be defined as either positive or negative, and it is important to understand the context which they are defined. For this, losses are defined as positive, representing the amount one could lose, or $\text{Loss} = |I_{\text{final}} - I_{\text{initial}}|$

5 Limitations

There are some problems with this simulation, despite its benefits of useful models and learning about investment fluctuation. Realistically, if in this scenario someone was using this model to try and seriously make money on the market, they would not be happy with these results. The final investments are extremely variable, and a majority of the investments make little to no to lose money.

5.1 Limitations of The Normal Model

The Normal Model Simulation provides data that is easy to work with and can be altered and studied very easily. However, some problems arise with it. Returns are not necessarily normally distributed. Although simulating them as normally distributed still provides a good understanding of the randomness of the financial market, returns are much harder to predict. This model also picks the returns from a set probability model, with mean expected return and standard deviation volatility. However, expected return and volatility change all the time. If you take a slightly different data set to calculate these statistics, a completely different result would be calculated and this affects the simulation.

As shown before, investments are a product of the set of its returns over a time period, once again given by the equation

$$I_{\text{final}} = I_{\text{initial}} \prod_{i=1}^{n-1} (1 + r_i)$$

While this can easily be fixed by taking the natural log of this equation, again shown before, it is something to keep in mind.

5.2 Limitations of The Historical Bootstrapping Model

The historical bootstrapping model is much less banking on a mathematical model to calculate returns, however it is negatively depending on some things. While measures of central tendency and spread help generalize a set of data that allows a normal model created from these statistics have less fluctuation from outliers, the historical bootstrapping model experiences none of this. Picking random returns from a set can lead to outliers having more place in the simulation than they actually would. So the data set that these returns are being picked from affects the historical bootstrapping model more than the normal model.

Looking at the set the returns are picked from, the past 10 years of daily returns, returns span a very different range of values. This is both good and bad. Companies like Tesla that experienced a large increase in stock prices will have much more variable returns, compared to companies like Apple and Microsoft that did not increase as much.